Individual Roles, Responsibilities and Project Explanation

Team Project Explanation

Here, we explained the complete working and implementation of the Smart Multi-Control Robotic Car project. We coordinated the hardware setup, Arduino programming, Bluetooth communication, testing, and final integration of all project modules.

We explained that the project is developed using Arduino Uno, L293D motor driver, HC-05 Bluetooth module, BO motors, ultrasonic sensor, servo motor, and a custom mobile application developed using MIT App Inventor.

We demonstrated how the MIT App sends Bluetooth commands to the Arduino Uno for controlling the robotic car movement such as forward, backward, left, right, and stop operations. We also explained the smart route buttons including Home, Hospital, Office, and School.

Further, we explained the automatic obstacle detection and smart navigation system using the ultrasonic sensor and servo motor scanning mechanism. The robotic car automatically detects nearby obstacles and avoids collisions during autonomous movement.

Finally, we presented the complete workflow, testing process, applications, advantages, and future scope of the project including Wi-Fi control, GPS navigation, camera integration, and AI-based automation systems.

Below, the roles and responsibilities were distributed among the team members according to their contributions in the project development, hardware integration, mobile application design, sensor automation, testing, and project presentation activities.

Bhagyashree

Role

MIT App & Website Developer

Responsibilities

- Developed the mobile application using MIT App Inventor
- Designed Bluetooth connection interface
- Created movement control buttons
- Developed route navigation buttons
- Added speed control interface
- Designed custom project information website using HTML, CSS, and JavaScript
- Tested app communication with Arduino system

Project Explanation

Bhagyashree explains how the MIT App controls the robotic car through Bluetooth communication. She demonstrates the forward, backward, left, right, and stop controls along with route buttons such as Home, Hospital, Office, and School. She also explains the frontend website developed for project presentation and information display using HTML, CSS, and JavaScript.

MIT App Logic Explanation

Bluetooth Connection Logic

When user clicks Bluetooth button, the app shows available Bluetooth devices. After selecting HC-05, the app connects and displays:

- Connected
  or
- Not Connected

Movement Control Logic

Each button sends a command to Arduino through Bluetooth.

Button	Command	Action
Forward	F	Car moves forward
Left	L	Car turns left
Backward	B	Car moves backward
Right	R	Car turns right
Stop	S	Car stops

Route Button Logic

Button	Command	Action
Home	1	Runs Home route
Hospital	2	Runs Hospital route
Office	3	Runs Office route
School	4	Runs School route
Auto Mode	5	Starts automatic navigation

Screen Navigation Logic

Button	Function
Button10	Opens Main Control Screen
Button11	Opens About Project Screen

Overall Working

The MIT App sends commands through Bluetooth to the Arduino Uno. Arduino receives the command and controls the motors and robotic car movement accordingly.

Shivangouda

Role

Hardware & Circuit Integration Engineer

Responsibilities

- Assembled the robotic car chassis
- Connected BO motors and wheels
- Integrated Arduino Uno and L293D motor driver
- Managed wiring and power supply setup
- Installed Bluetooth module and tested connections
- Performed motor movement testing

Project Explanation

I, Shivangouda, explained the complete hardware setup and working mechanism of the Smart Multi-Control Robotic Car project. I described the role of each hardware component used in the system and how all components are connected together to achieve robotic vehicle movement.

I explained that the Arduino Uno acts as the main controller of the project. It receives commands from the Bluetooth module and processes them to control the movement of the robotic car.

I also explained the working of the L293D motor driver, which is used to control the direction and movement of the BO motors. The motor driver receives signals from Arduino and supplies required power to the motors for forward, backward, left, and right movement.

The HC-05 Bluetooth module is used for wireless communication between the MIT App and Arduino Uno. When the user presses buttons in the mobile application, commands are sent through Bluetooth to control the robotic car.

I also explained the BO motors and wheel assembly used for robotic movement. I demonstrated how motor rotation helps the car move in different directions.

I further explained the battery and power supply connections, which provide necessary power to Arduino, motor driver, Bluetooth module, and motors for smooth operation of the project.

Finally, I demonstrated the practical working of the robotic car by showing movement control, Bluetooth connectivity, and motor operation using the MIT App interface.

Chetana

Role

Sensor & Automation Developer

Responsibilities

- Integrated ultrasonic sensor system
- Implemented obstacle detection mechanism
- Worked on automatic navigation logic
- Tested distance sensing functionality
- Developed smart obstacle avoidance features
- Assisted in route automation testing

Project Explanation

I, Chetana, explained the working principle of the ultrasonic sensor and automatic obstacle detection system used in the Smart Multi-Control Robotic Car project. I demonstrated how the ultrasonic sensor detects nearby obstacles by measuring the distance between the robotic car and objects in front of it.

I also explained how the robotic car automatically stops when an obstacle is detected within a specific distance to avoid collisions. The sensor continuously sends distance data to the Arduino Uno, which processes the information and controls the movement of the robotic car accordingly.

I further demonstrated the smart navigation and automatic obstacle avoidance system, where the servo motor scans left and right directions to identify a safer path for movement during autonomous mode.

Finally, I explained how the automatic safety system improves the efficiency, intelligence, and safe movement of the robotic vehicle during operation.

Supriya

Role

Documentation & Presentation Coordinator

Responsibilities

- Prepared project documentation
- Created presentation materials and posters
- Managed project testing and result recording
- Documented applications and future scope
- Coordinated final project presentation

Project Explanation

I, Supriya, explained the project objectives, applications, advantages, future scope, and overall workflow of the Smart Multi-Control Robotic Car project. I presented the complete project documentation, testing results, and overall system performance during the project demonstration.

I explained that the main objective of the project is to develop a smart robotic vehicle capable of wireless Bluetooth control, automatic navigation, and obstacle detection using Arduino Uno and MIT App control.

I also described the various applications of the project such as smart robotic systems, obstacle avoidance vehicles, automation projects, and educational robotics learning.

Further, I explained the advantages of the system including wireless control, smart movement, user-friendly mobile application interface, and automatic safety features.

I also discussed the future scope of the project, where the robotic car can be upgraded using Wi-Fi control, GPS navigation, live camera streaming, voice control, and AI-based autonomous systems for more advanced smart vehicle applications.

Finally, I presented the testing process, project workflow, and final project outcomes during the demonstration.